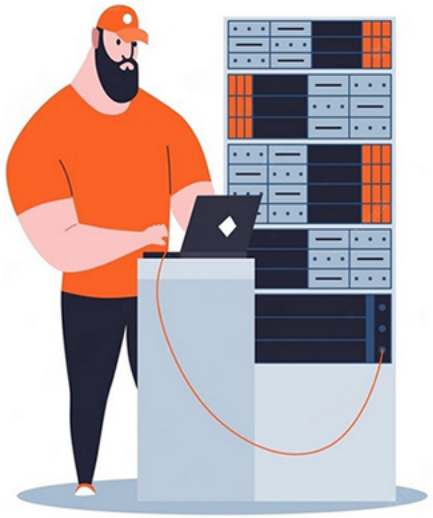
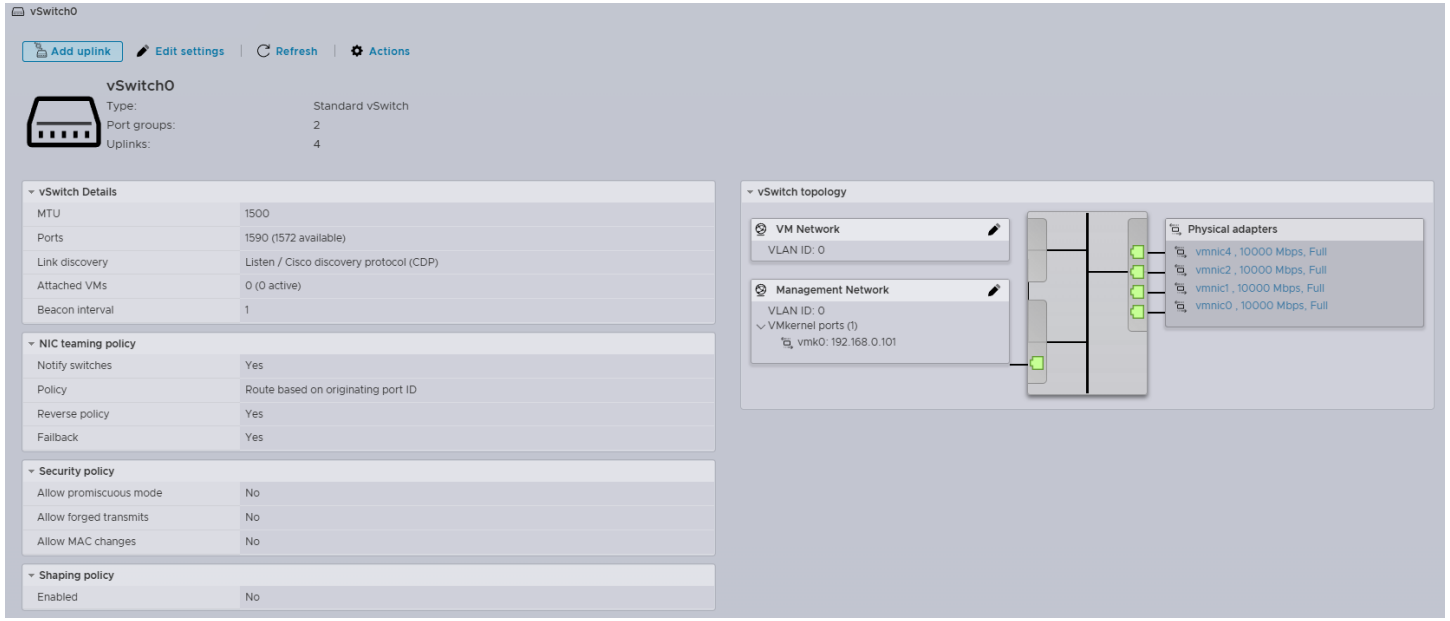
AGENDA

About vSphere Standard Switch (vSS).
About Physical Network Interface Cards (pNICs).
About Port Group in vSS.
About Virtual Machine Port Group.
About VMKernal Port Group.
About vLan’s.





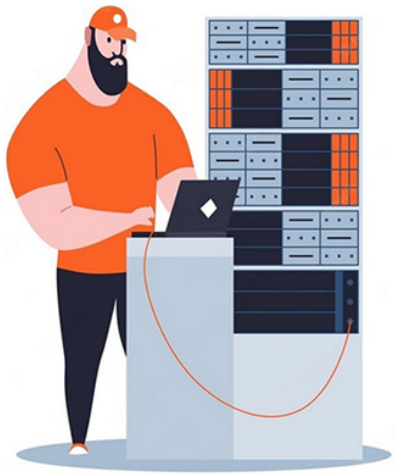
VMware ESXi
HYPERVISOR

About vSphere Standard Switch (vSS)

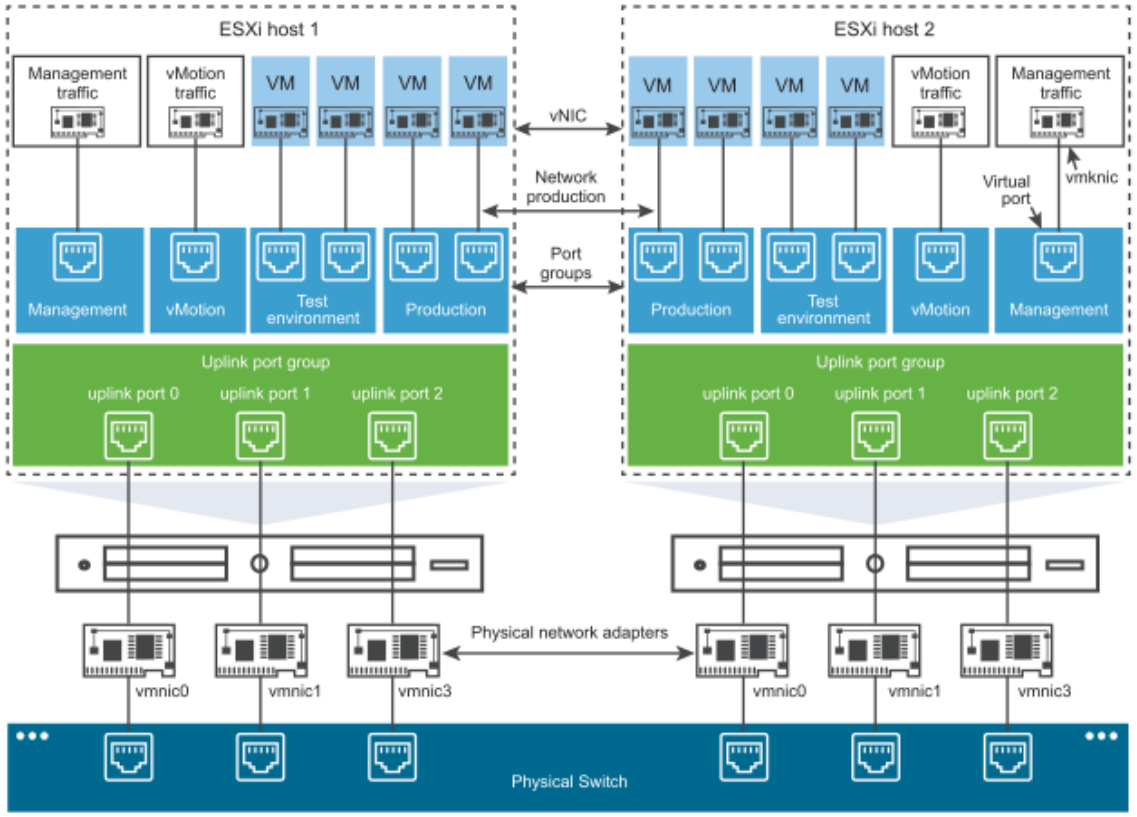


About vSphere Standard Switch

To provide network connectivity to hosts and virtual machines, you connect the physical NICs of the hosts to uplink ports on the standard switch. Virtual machines have network adapters (vNICs) that you connect to port groups on the standard switch. Every port group can use one or more physical NICs to handle their network traffic. If a port group does not have a physical NIC connected to it, virtual machines on the same port group can only communicate with each other but not with the external network.



vSphere Standard Switch



vSphere Standard Switch

About vSphere Standard Switch



Physical Network Interface Cards (pNICs)

- These are the physical network interfaces on the ESXi host. They are used to connect the host to the outside world, such as to the physical network infrastructure (switches, routers, etc.).
- ESXi supports multiple pNICs, which can be teamed together for redundancy (NIC teaming) or load balancing.

Port Groups

- Port groups are used to categorize virtual machine network traffic. A port group defines a set of network configuration settings for virtual machines (such as VLAN ID, security policies, etc.).
- VMs that are connected to a port group share the same networking configuration and can communicate with each other, as well as external networks, depending on the vSwitch and physical NIC configuration.

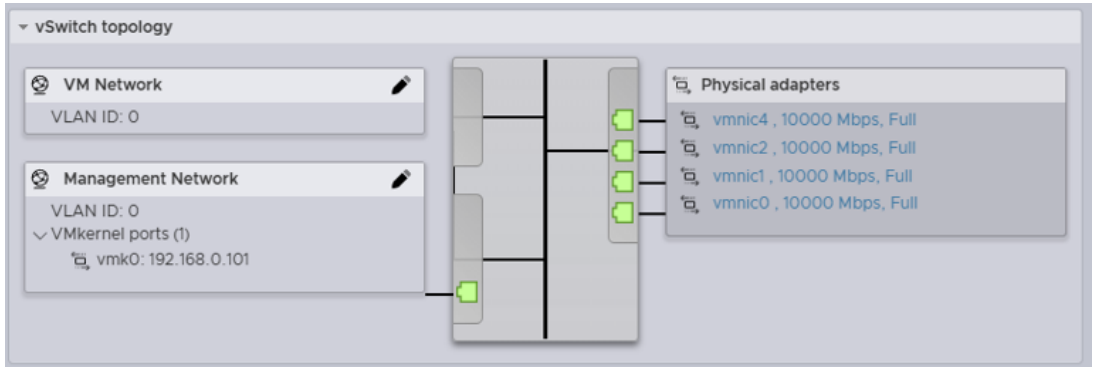
Virtual Machine Port Group

- Virtual Machine Ports are used for connecting virtual machine to physical host and vm's can communicate each other with in same virtual machine port group.

VMkernel Port Group

- VMkernel ports are used for host-level network services such as vMotion, iSCSI, NFS, vSphere HA, and other ESXi management functions. These are special network interfaces that are not associated with virtual machines but with ESXi itself.
- Each VMkernel port is associated with a specific network interface on the ESXi host, and it can have its own IP address and network configuration.





About vSphere Standard Switch



VLANs (Virtual LANs)

- VLANs are used to segment network traffic into different logical networks, even on the same physical network.
- ESXi can tag outgoing traffic from VMs based on the VLAN ID configured on the port group, and it can also be configured to trunk multiple VLANs from the physical NICs to allow for communication across VLANs.

